Chaeun Park

chaeunnara@snu.ac.kr Seoul, South Korea chaephymath

Education

Seoul National University, Physics and Mathematical Sciences (Double Major) — Seoul, South Korea
- College of Natural Sciences — 2022 – Aug 2027

Research Interests

Nonlinear PDEs and Fluids
Existence, regularity, and nonlinear dynamics of fluid equations, particularly the Euler equations, with attention to the effects of boundaries and singular geometry.

Hyperbolic Geometry and Geometric Group Theory
Discrete group actions on hyperbolic spaces, especially Fuchsian groups, and how group-theoretic and representation-theoretic properties are reflected in geometry, topology, and dynamics.

Quantum Field Theory and Gravitation
The relation between quantum field theory and spacetime geometry, from conserved quantities such as Komar integrals and ADM mass to holographic duality in AdS/CFT. I am also interested in how black-hole physics connects quantum chaos, random matrix theory, and SYK models.

Research & Project Experience

Computational Study of Pseudomodular Fuchsian Groups, Research Intern, advised by Prof. Gye-Seon Lee — SNU, Seoul, South Korea, Mar 2026 – Aug 2026
- Studied constructions of pseudomodular Fuchsian groups through Long–Reid's "Pseudomodular Surfaces," focusing on the relationship between rational cusp sets, arithmeticity, and commensurability.
- Developed Python/SymPy computations to reproduce trace-based arithmeticity tests, detect rational fixed points of hyperbolic elements, and construct killer-interval coverings.
- Reproduced computational certificates for the pseudomodularity of $\Delta(2/5, 4)$ and $\Delta(3/7, 4)$, with an appendix-guided verification for $\Delta(3/11, 4)$; documented the work in an internship report.

Local Well-Posedness of Incompressible Euler Equations on Singular Domains, Research Intern, advised by Prof. Injee Jeong (now at KIAS) — SNU, Seoul, South Korea, Mar 2025 – Aug 2025
- Investigated local well-posedness of incompressible Euler equations in Sobolev spaces via velocity–pressure methods and energy estimates.
- Studied energy and pressure estimates in whole-space, half-space, and corner domains, including boundary compatibility and elliptic regularity.
- Prepared a research report and presented the work.

SEGL (SNU Experimental Geometry Lab) Project, Project Member, advised by Prof. Gye-Seon Lee — SNU, Seoul, South Korea, 2023
- Studied discrete hyperbolic reflection groups in the SEGL (SNU Experimental Geometry Lab) project, drawing on Hohlweg–Labbé–Ripoll's work on infinite Coxeter groups.
- Contributed theoretical analysis and hand calculations to a collaborative Python visualization of limit sets.

Presentations & Posters

The Power of Symmetry: From Rotations to Particles, Forces, and Gravity — CERN, Geneva, Switzerland, July 2026
Joint presentation with Drim Choi and Seokwon Lee at the CERN–CKC Summer Student Programme 2026. Presented Part III, "Massless Spin-2 and Einstein Gravity," on how gauge invariance and the self-consistent coupling of a massless spin-2 field lead from Fierz–Pauli theory to Einstein gravity.

Parabolic Krylov–Safonov Harnack Inequality — SNU, Seoul, South Korea, June 2026
Final presentation for Prof. Sun-Sig Byun's special-topics course on regularity methods for nonlinear elliptic and parabolic equations. Based on Beomjun Choi's notes, covered parabolic ABP estimates, measure and propagation lemmas, Vitali covering, level-set decay, weak and full Harnack inequalities, and Hölder regularity.

Euler Equations on 2D Singular Domains — SNU, Seoul, South Korea, Dec 2025
Final presentation for Prof. Sun-Sig Byun's special-topics course on regularity theory for partial differential equations. Reviewed classical Yudovich theory and its extensions and limitations in nonsmooth domains with corners, emphasizing log-Lipschitz estimates, Lagrangian flows, and the Osgood criterion.

PIGE Analysis of $^{12}C(p,\gamma)^{13}N$ and $^{27}Al(p,p'\gamma)^{27}Al$, $^{27}Al(p,\alpha\gamma)^{24}Mg$ Reactions using Pelletron Accelerator — SNU, Seoul, South Korea, Dec 2024
Poster presentation with Haeun Chung at the 2024 Undergraduate Research Presentation, based on experimental nuclear physics research conducted during Nishina School 2024.

Honors, Awards & Scholarships

Hyunsong Scholarship — 2023 – 2025
Awarded by the Hyunsong Educational & Cultural Foundation every semester from 2023 to 2025 (six semesters).

Excellence Award, 2024 Undergraduate Research Presentation — SNU, Seoul, South Korea, Dec 2024
Awarded by the Department of Physics and Astronomy for the poster "PIGE Analysis of $^{12}C(p,\gamma)^{13}N$ and $^{27}Al(p,p'\gamma)^{27}Al$, $^{27}Al(p,\alpha\gamma)^{24}Mg$ Reactions using Pelletron Accelerator," based on research conducted during Nishina School 2024.

Teaching & Mentoring

Topology Mentoring, Mentor — Seoul, South Korea, Apr 2026 – Aug 2026
Private, in-person undergraduate mentoring based on Chapters 2–8 of Munkres' Topology.

Algebra Mentoring, Mentor — Seoul, South Korea, Jan 2024 – Feb 2024
Private, in-person mentoring for two students, covering selected topics from Chapters 1–14 of Dummit and Foote's Abstract Algebra.

Academic Study Groups

Advanced Quantum Field Theory, Lead Organizer and Presenter — SNU, Seoul, South Korea, Jan 2026 – present
Co-leading an advanced, year-long study of quantum field theory with a small group of core presenters. Topics are organized primarily around Ramond, with selected material from Schwartz, Srednicki, and other references.

Introductory Quantum Field Theory, Presenter — SNU, Seoul, South Korea, Jan 2025 – Feb 2025
Participated in a rotating seminar-style introduction to quantum field theory based on Ramond's Field Theory: A Modern Primer, with Professor Seok Kim invited as faculty mentor.

Compact Lie Group Representation Theory, Presenter — SNU, Seoul, South Korea, Sept 2024 – Dec 2024
Participated in a rotating seminar based on Representations of Compact Lie Groups.

Clubs

SQRT (SNU Quantum Research Team), Member — SNU, Seoul, South Korea — Sept 2022 – Jan 2025
Participated in club study sessions on quantum computing, including fundamentals of quantum information theory and superconducting quantum circuits. Recognized as an SQRT alumnus (certificate issued February 26, 2025).

Convergence — Mathematics Academic Club, Member — SNU, Seoul, South Korea — 2023

Mathematics Schools & Workshops

The 6th International Undergraduate Mathematics Summer School — SNU, Seoul, South Korea
Participant — 2026-07-27 to 2026-08-07

The 12th Korea PDE Winter School — POSTECH, Pohang, South Korea
Participant — 2026-01-19 to 2026-01-23

The 3rd International Undergraduate Mathematics Summer School — SNU, Seoul, South Korea
Participant — 2023-08-07 to 2023-08-18

10-10 Summer School on Mathematics of Deep Learning and AI — SNU, Seoul, South Korea
Participant — 2023-08-07 to 2023-08-08

Physics Schools & Workshops

CERN-Korean Summer Student Program — CERN, Geneva, Switzerland
Participant — 2026-06-29 to 2026-07-11

SNU-APCTP Winter School on Fundamental Physics — SNU, Seoul, South Korea
Participant — 2026-02-02 to 2026-02-06

Nishina School 2024 — RIKEN, Japan
Participant — 2024-07-25 to 2024-08-02

Languages/Skills

- Korean, English
- Python, Mathematica